

Notes de Surfaces de Riemann

GitHub Repository: [Oxke/appunti/SurfRiemann](https://github.com/Oxke/appunti/SurfRiemann)

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Chapter 1

Boh

Chapter 2

Quotients

Soit X une Riemann Surface. $\text{Aut}(X) = \{\varphi : X \rightarrow X \text{ holomorphic}\}$. Si on prends $\Gamma \subseteq \text{Aut}(X)$ subgroup, can X/Γ the set of orbits be equipped with a complex structure?

Definition 1

1. The action of Γ on X is **proper** $\iff$ for all compact $K \subseteq X$, then $\{\gamma \in \Gamma : \gamma(K) \cap K \neq \emptyset\}$ is finite.
2. The action is **free** if there are no fixed points, i.e.

$$\forall \gamma \in \Gamma \setminus \{\text{Id}\}, \forall x \in X, \gamma x \neq x$$

Proposition 2. *If the action of Γ on X is proper, then X/Γ is Hausdorff. If it is also free, then $\pi : X \rightarrow X/\Gamma$ is a topological covering, i.e. $\forall a \in X$ there exists $U_a \ni a$ open such that $\pi|_{U_a}$ is a homeomorphism and $\pi^{-1}(\pi(U_a)) \coprod_{\gamma \in \Gamma} \gamma(U_a)$*

Theorem 3: Complex structure on X/Γ

Let X be a Riemann Surface. Assume $\Gamma \subseteq \text{Aut}(X)$ has a proper action on X . Then there is a unique complex structures on the quotients such that $\pi : X \rightarrow X/\Gamma$ is holomorphic.

proof if also free action. Assume the action to be also free. For all $p \in X$, choose a chart on X $(U_p, \varphi_p) \ni p$ such that $\pi|_{U_p}$ is a homeomorphism and $\pi^{-1}(\pi(U_p)) = \bigcup_{\gamma \in \Gamma} \gamma(U_p)$ and call $\pi(U_p) =: V_{\pi(p)}$. Then $(V_{\pi(p)}, \varphi_p \circ \pi^{-1}|_{V_{\pi(p)}})$ is an atlas on X/Γ .

If $V_{\pi(p)} \cap V_{\pi(p')} \neq \emptyset$ then

$$\psi_{\pi(p)} \circ \psi_{\pi(p')}^{-1} = (\varphi_p \circ \pi^{-1}|_{V_{\pi(p)}}) \circ (\pi|_{U_{p'}} \circ \varphi_{p'}^{-1})$$

and the middle two composed make a function $\gamma_{p,p'}$ holomorphic. $\square$

If instead there are fixed points then we set $\Gamma_x = \{\gamma \in \Gamma : \gamma x = x\}$ the stabilizer of x .

Proposition 4. Γ_x is a finite subgroup.

Proposition 5. $\{x \in X : \Gamma_x \neq \{\text{Id}\}\}$ is a discrete set.

Then to prove the theorem 2 we will consider that the action is proper and the fixed points for $\gamma \neq \text{Id}$ are isolated.

Proposition 6. *If $x \in F$, one can find $U \ni x$ such that*

1. $\gamma(U) \cap U = \emptyset \quad \forall \gamma \notin \Gamma_x$
2. $\Gamma_x(U) \subseteq U$
3. $\pi(U) \cong U/\Gamma_x$

Pick a chart (U, φ) and define a map

$$\begin{aligned} \lambda : \Gamma_x &\longrightarrow \mathbb{C}^* \\ \gamma &\longmapsto \lambda(\gamma) = (\varphi \circ \gamma \circ \varphi^{-1})'(0) \end{aligned}$$

then λ is a group homomorphism and it is injective, indeed if $\lambda(\gamma) = 1$, then $\varphi \circ \gamma \circ \varphi^{-1}(z) = z + \alpha z^k + O(z^{k+1})$. Calling $n = |\Gamma_x|$ and observing that $\gamma^n = \text{Id}$, we have

$$\varphi \circ \gamma^n \circ \varphi^{-1}(z) = z = z + n\alpha z^k + O(z^{k+1})$$

which means that $\alpha = 0$. Hence, we have that finite $\Gamma_x \cong \lambda(\Gamma_x) \leq \mathbb{C}^*$ hence Γ_x is cyclic.

Set $\omega_x(p) = \frac{1}{n} \sum_{\gamma \in \Gamma_x} \lambda(\gamma)^{-1} \cdot \varphi(\gamma(p))$. We have $\omega_x(x) = 0$ and ω_x is indeed a chart (by shrinking U).

Proposition 7. $\forall \gamma \in \Gamma_x, \omega_x(\gamma(p)) = \lambda(\gamma) \cdot \omega_x(p)$

Since Γ_x is cyclic, we can pick a generator γ_0 of Γ_x such that $\lambda(\gamma_0) = e^{\frac{2i\pi}{n}}$. In conclusion, in the chart U/Γ_x is $\omega_x(U)/\langle z \mapsto e^{\frac{2i\pi}{n}} z \rangle$. We will call $U_n := \langle z \mapsto e^{\frac{2i\pi}{n}} z \rangle$

Then $p \mapsto \omega_x^n(p)$ is a chart on U/Γ_x . We would only need to check that it makes an atlas. (v. Miranda Ch. 3 p 75).

Exercise 8

Let X, Y be Hausdorff topological spaces and $\pi : X \rightarrow Y$ a local homeomorphism. If Y has a complex atlas, then there exists a complex atlas on X such that π is holomorphic.

2.0.1 Example of Quotients

Tori: Let w_1, w_2 be two $\mathbb{R}$ -independent and set $\Lambda = \mathbb{Z}w_1 \oplus \mathbb{Z}w_2$. Then Λ acts by translation on $\mathbb{C}$. Then Λ is discrete and the action is proper.

Then $X = \mathbb{C}/\Lambda$ is called a **torus**, a compact Riemann surface and it's not homeomorphic to $\widehat{\mathbb{C}}$

Modular Surface: Let $\mathbb{H} = \{z \in \mathbb{C} : \text{Im}z > 0\}$. Set $\text{PSL}_2(\mathbb{Z}) \cong \Gamma := \{h(z) = \frac{az+b}{cz+d} : ad-bc=1, a, b, c, d \in \mathbb{Z}\}$. Then Γ leaves $\mathbb{H}$ invariant. Γ acts properly on $\mathbb{H}$.

Γ has fixed points. For example $(S : z \mapsto -\frac{1}{z}) \in \Gamma$ and $S(i) = i$. Setting $(T : z \mapsto z+1) \in \Gamma$ then $ST(z) = -\frac{1}{z+1}$ and $ST(e^{\frac{2i\pi}{3}}) = e^{\frac{2i\pi}{3}}$

Proposition 9. $F = (\Gamma \cdot i) \cup \left(\Gamma \cdot e^{\frac{2i\pi}{3}} \right)$

Chapter 3

Algebraic Curves

3.1 Main Tools

The main tool we will use in this chapter is the following, and will be stated and proved for polynomials, even though it would be more general.

Theorem 10: Implicit function theorem

Let $f \in \mathbb{C}[z, w]$. Consider $\mathcal{C} = \{(x, y) \in \mathbb{C}^2 : f(x, y) = 0\}$. Assume then $(z_0, w_0) \in \mathcal{C}$ and $\frac{\partial f}{\partial w}(z_0, w_0) \neq 0$. Then there exists $U \ni z_0$ and $V \ni w_0$ open sets and an holomorphic map $g : U \rightarrow \mathbb{C}$ holomorphic such that

$$\mathcal{C} \cap U \times V = \{(z, g(z)) : z \in U\}$$

Proof. Use first the C^∞ -implicit function theorem. Write then $f(z, w) = \sum \alpha_{l,j} z^l w^j$ and we have

$$\partial_w f = \sum \alpha_{l,j} z^l j w^{j-1} dw = \frac{\partial f}{\partial w} dw \implies \partial_w f(z_0, w_0) = \underbrace{\frac{\partial f}{\partial w}(z_0, w_0)}_{\neq 0} dw$$

hence $\partial_w f(z_0, w_0)$ is invertible and thus there exists U, V open sets and $g : U \rightarrow \mathbb{C}$, $g \in C^\infty$ such that $\mathcal{C} \cap (U \times V) = \{(z, g(z)) : z \in U\}$.

We just now need to show that g is holomorphic. Set $F : U \rightarrow \mathbb{C}$ given by $F(z) := f(z, g(z)) = 0$ on U . Then

$$d_z F = 0 \iff \frac{\partial F}{\partial z} dz + \frac{\partial F}{\partial \bar{z}} \bar{z} = 0$$

Hence

$$\frac{\partial F}{\partial \bar{z}} = 0 = \frac{\partial f}{\partial w}(z, g(z)) \cdot \frac{\partial g}{\partial \bar{z}}(z) \implies \frac{\partial g}{\partial \bar{z}} \equiv 0 \text{ near } z_0$$

hence g is holomorphic. □

Theorem 11

Let $f \in \mathbb{C}[z, w]$ such that

1. $\forall (z, w) \in \mathcal{C}, \left(\frac{\partial f}{\partial z}, \frac{\partial f}{\partial w} \right) \neq (0, 0)$
2. f is irreducible in $\mathbb{C}[z, w]$

Then there exists a complex structure on $\mathcal{C}$ such that the two projections $p_1, p_2 : \mathcal{C} \rightarrow \mathbb{C}$, given by $(z, w) \mapsto z$ and $(z, w) \mapsto w$ respectively, are holomorphic.

Remark. The two regularity requisites ensure that (1.) we can use the implicit function theorem and (2.) that $\mathcal{C}$ is connected. This second result would require a deeper explanation coming from algebraic geometry.

Proof. By (1.) if $(z_0, w_0) \in \mathcal{C}$ then one of the two derivatives is nonzero, let's assume $\frac{\partial f}{\partial w} \neq 0$. Using the IFT (theorem 3.1) we know that

$$\underbrace{\mathcal{C} \cap (U \times V)}_{\mathcal{O}_0} = \{(z, g(z)) : z \in U\} \text{ with } g \text{ holomorphic}$$

Then define $\varphi_0 : \mathcal{O}_0 \rightarrow \mathbb{C}$ by $\varphi_0(z, w) = z = p_1(z)$. Then $(\mathcal{O}_0, \varphi_0)$ is a chart. Similarly, if $\frac{\partial f}{\partial z} \neq 0$ then the chart will be $(\mathcal{O}_0, \varphi_0 = p_2)$.

We say that

$$\bigcup_{(z_0, w_0) \in \mathcal{C}} (\mathcal{O}_0, \varphi_0) \text{ is an atlas}$$

indeed, if $\mathcal{O} \cap \mathcal{O}' \neq \emptyset$, then $\varphi' \circ \varphi^{-1}(z) = g(z)$ holomorphic.

Finally, note that p_1 and p_2 are holomorphic because they are charts. $\square$

3.2 Projective Curves

Let $F \in \mathbb{C}[x, y, z]$ such that F is homogenous, i.e. $\forall \lambda \in \mathbb{C}, F(\lambda x, \lambda y, \lambda z) = \lambda^d F(x, y, z)$ with $d \in \mathbb{Z}_{\geq 0}$. Then we define

$$\mathcal{C} = \{[x, y, z] \in \mathbb{P}^2 \mathbb{C} : F(x, y, z) = 0\}$$

which makes sense because F is homogenous.

Definition 12: Smooth

A projective curve $\mathcal{C}$ is said to be **smooth** if $\forall [x, y, z] \in \mathcal{C}, dF \neq 0$

Theorem 13

Assume $\mathcal{C}$ is *smooth*. Then there exists a complex structure on $\mathcal{C}$ such that $\mathcal{C}$ is a compact connected Riemann Surface.

Proof. v. [?] p. 15 $\square$

Exercise 14

$\mathcal{C} \subseteq \mathbb{P}^2 \mathbb{C}$ is smooth $\iff \forall i = 1, 2, 3, U_i \cap \mathcal{C}$ is *affine smooth*.

Example 15. Let $F(x, y, z) = x^d + y^d + z^d$. Then F is homogenous of degree d , hence it defines a projective curve $\mathcal{C}$. Moreover, $\mathcal{C}$ is smooth, since $\frac{\partial F}{\partial \star} = d\star^{d-1}$, with $\star = x, y, z$.

Remark. If $\mathcal{C} = \{f(u, v) = 0\} \subseteq \mathbb{C}^2$ is affine, one can consider $F(x, y, z) = z^d f(\frac{x}{z}, \frac{y}{z})$ and $\widehat{\mathcal{C}} = \{[x, y, z] \in \mathbb{P}^2 \mathbb{C} : F(x, y, z) = 0\}$ and $\widehat{\mathcal{C}} \cap U_3 = \mathcal{C}$. If $\mathcal{C}$ is smooth, then $\widehat{\mathcal{C}}$ is in general non smooth.

Exercise 16

Consider $\mathcal{C} = \{(x, y) \in \mathbb{C}^2 : y^2 = P(x)\}$ where $P \in \mathbb{C}[X]$ non constant and P has simple roots.

1. Check that $\mathcal{C}$ is smooth

2. Check that $\widehat{\mathcal{C}}$ is smooth $\iff d \leq 3$

Chapter 4

Elliptic Curves

4.1 Tori

Let $\Lambda = \mathbb{Z}w_1 + \mathbb{Z}w_2$, with $w_1, w_2 \in \mathbb{C}$ $\mathbb{R}$ -independent.

Proposition 17. *For all $\alpha > 2$, the sum $\sum_{0 \neq w \in \Lambda} \frac{1}{\|w\|^\alpha}$ converges*

short proof. Set $N(R) = |\{w \in \Lambda : \|w\| \leq R\}|$. Then $N(R) = o(R^2)$. Then

$$\sum_{0 \neq w \in \Lambda} \frac{1}{\|w\|^\alpha} = \int_{0+}^{\infty} \frac{dN(r)}{r^\alpha} = O(1) + \underbrace{\int_{0+}^{\infty} \frac{\alpha}{u^{\alpha+1}} N(u) du}_{\text{convergent} \iff \alpha > 2}$$

□

Consider $\mathbb{T} = \mathbb{C}/\Lambda$. Then the Weierstrass function, $\forall z \in \mathbb{C} \setminus \Lambda$, is defined by

$$\mathfrak{p}(z) = \frac{1}{z^2} + \sum_{0 \neq w \in \Lambda} \left(\frac{1}{(z-w)^2} - \frac{1}{w^2} \right)$$

Proposition 18 (Properties of Weierstrass functions).

- *On compact subsets $K \subseteq \mathbb{C} \setminus \Lambda$ the formula is convergent, since $\left| \frac{1}{(z-w)^2} - \frac{1}{w^2} \right| = O_K\left(\frac{1}{\|w\|^3}\right)$.*
- *$\mathfrak{p}$ is holomorphic on $\mathbb{C} \setminus \Lambda$ and meromorphic on $\mathbb{C}$, with poles exactly on Λ .*
- *$\mathfrak{p}$ is Λ -invariant. We can look at $\mathfrak{p}'$.*

$$\mathfrak{p}'(z) = -\frac{2}{z^3} + \sum_{w \neq 0} -\frac{2}{(z-w)^3} = -\sum_{w \in \Lambda} \frac{2}{(z-w)^3}$$

and clearly $\mathfrak{p}'(z + w_0) = \mathfrak{p}'(z)$ for all $w_0 \in \Lambda$.

Now fix $w_0 \in \Lambda$ and look at $F : \mathfrak{p}(z + w_0) - \mathfrak{p}(z)$ on $\mathbb{C} \setminus \Lambda$. Then $F'(z) = 0$ on $\mathbb{C} \setminus \Lambda$ which is connected. We get that $F(z) = C_{w_0}$ some constant. To prove that $C_{w_0} = 0$ we can observe that $\mathfrak{p}(z) = \mathfrak{p}(-z) \forall z \in \mathbb{C} \setminus \Lambda$

Note (ation). For $p > 2$, we set

$$S_p := \sum_{w \in \Lambda \setminus \{0\}} \frac{1}{w^p}$$

hence near $z = 0$, the Laurent expansion of $\mathfrak{p}$ gives

$$\begin{aligned}\mathfrak{p}(z) &= \frac{1}{z^2} + \sum_{p=1}^{\infty} (2p+1)S_{2p+2}z^{2p} = \frac{1}{z^2} + a_1z^2 + a_2z^4 + O(z^6) \\ a_1 &= 3S_4 \quad a_2 = 5S_6 \quad ; \quad \mathfrak{p}'(z) = -\frac{2}{z^3} + 2a_1z + 4a_2z^3 + O(z^5) =\end{aligned}$$

Hence we get that

$$(\mathfrak{p}'(z))^2 = P(\mathfrak{p}(z)) \tag{4.1.1}$$

with P a polynomial of degree 3.

Now we can define

$$\begin{aligned}\varphi : X &\longrightarrow \mathbb{P}^2\mathbb{C} \\ z &\longmapsto \varphi(z) = \begin{cases} [\mathfrak{p}(z), \mathfrak{p}'(z), 1] & \text{if } z \neq \pi(0) \\ [0, 1, 0] & \text{if } z = \pi(0) \end{cases}\end{aligned}$$

Theorem 19

$\varphi : X \rightarrow \widehat{\mathcal{C}}$ is biholomorphic, with $\mathcal{C} = \{(x, y) \in \mathbb{C}^2 : y^2 = P(x)\}$ and $\widehat{\mathcal{C}}$ is the projectification of $\mathcal{C}$ in $\mathbb{P}^2\mathbb{C}$

Proof. We know that $\frac{\mathfrak{p}(z)}{\mathfrak{p}'(z)} \rightarrow 0$ as $z \rightarrow 0$. Hence for $z \sim 0$

$$[\mathfrak{p}(z), \mathfrak{p}'(z), 1] = \left[\frac{\mathfrak{p}(z)}{\mathfrak{p}'(z)}, 1, \frac{1}{\mathfrak{p}'(z)} \right] \rightarrow [0, 1, 0]$$

as $z \rightarrow 0$ and we need to show that φ is injective, because then it is automatically surjective and biholomorphic.

Suppose now $z, z' \in \mathcal{F}$, $\mathfrak{p}(z) = \mathfrak{p}(z')$ and $\mathfrak{p}'(z) = \mathfrak{p}'(z')$.

- If $\mathfrak{p}'(z) = 0$, then $\{z, z'\} \subseteq \{z_1, z_2, z_3\}$, however if $z \neq z'$ then $\mathfrak{p}(z) \neq \mathfrak{p}(z')$, thus $z = z'$.
- If instead $\mathfrak{p}'(z) \neq 0$, then $z = -z' \bmod \Lambda$ and by the oddness of $\mathfrak{p}'$ we get that $\mathfrak{p}'(z) = \mathfrak{p}'(z') = 0$ which is absurd.

□

Theorem 20: Residue

Assume $f : \mathbb{C} \rightarrow \mathbb{C}$ is meromorphic and Λ -invariant.

Set $\mathcal{F} = [0, 1)w_1 + [0, 1)w_2$ and let $z_1, \dots, z_n$ be poles of f in $\mathcal{F}$. Then

$$\sum_{i=1}^n \text{Res}_{z_i}(f) = 0$$

Proof. Up to translating the fundamental domain slightly we can assume that there are no poles on $\partial\mathcal{F}$, then let's take the contour $\gamma = \partial\mathcal{F}$. We get that

$$\frac{1}{2\pi} \int_{\gamma} f(z) dz = \sum_{i=1}^n \text{Res}_{z_i}(f)$$

By the usual residue theorem. However f is Λ -periodic, hence “opposite sides” in $\int_{\gamma} f$ cancel out and thus we get the thesis. □

Remark. Applying the residue theorem to $\frac{f'}{f}$, we get that $Z(f) - P(f) = 0$, where $Z(f)$ is the number of zeros and $P(f)$ the number of poles inside $\mathcal{F}$.

Corollary 21 (Consequence 1). *$\mathfrak{p}'$ has 3 zeros in $\mathcal{F}$. Since $\mathfrak{p}$ is an even function, $\mathfrak{p}'$ is odd. Hence, since $\mathfrak{p}'$ is Λ -invariant, we get that it has three zeros:*

$$z_1 = \frac{w_1}{2} \quad ; \quad z_2 = \frac{w_2}{2} \quad ; \quad z_3 = \frac{w_1 + w_2}{2}$$

which thus are simple zeros.

Corollary 22 (Consequence 2). *For all $w \in \mathbb{C}$, $\mathfrak{p}(z) = w$ has two zeros in $\mathcal{F}$. If moreover $\mathfrak{p}'(z) \neq 0$, then there are two different solutions.*

In summary, if $w \notin \mathfrak{p}(\{z_1, z_2, z_3\})$ (not in the image of the ramification points), then $|\mathfrak{p}^{-1}(\{w\})| = 2$

Moreover, in this situation, if $\mathfrak{p}(z) = \mathfrak{p}'(z) = w$, then $z = -z' \pmod{\Lambda}$

Proposition 23 (Obvious Fact). *$\mathbb{C}/\Lambda$ is an abelian group.*

One can now transport via φ this group structure on $\mathcal{C}$. Call $\oplus$ the addition on $\mathcal{C}$, then $A_1 \oplus A_2 = \overline{A_3}$ as in the

Moreover, if $A_i = (\mathfrak{p}(w_i), \mathfrak{p}'(w_i)) \in \mathbb{C}^2$ then $A_1 \oplus A_2 = \overline{A_3} \iff w_1 + w_2 + w_3 = 0 \pmod{\Lambda}$

ideas:

- Write $y = ax + b$ for the equation of $\mathcal{D}$ in $\mathbb{C}^2$
- Consider $\psi(z) = \mathfrak{p}'(z) - a\mathfrak{p}(z) - b$
- w_1, w_2, w_3 are zeros of ψ in $\mathcal{F}$
- Apply **Abel's identity for elliptic functions:**

$$z_1 + \dots + z_n - p_1 - \dots - p_m = 0 \pmod{\Lambda}$$

where $z_1, \dots, z_n$ are the zeros and $p_1, \dots, p_m$ are the poles of ψ inside $\mathcal{F}$.

Chapter 5

Topology and Holomorphic maps

Lemma 24 (Normal form of holomorphic maps). *Let $f : X \rightarrow Y$ a non-constant holomorphic map. Fix $P \in X$. Then there exists $m_P \in \mathbb{Z}_{\geq 1}$ such that for all charts $(U_2, \varphi_2) \ni f(P)$, $\varphi(f(P)) = 0$. Then one can find a chart $(U_1, \varphi_1) \ni P$ with $\varphi_1(P) = 0$ such that near 0, we have*

$$\varphi_2 \circ f \circ \varphi_1^{-1}(z) = z^{m_P}$$

Proof. We start with two charts $(U_1, \varphi_1) \ni P$ and $(U_2, \varphi_2) \ni f(P)$ such that $\varphi_1(P) = 0$ and $\varphi_2(f(P)) = 0$.

Near 0, we have that $\varphi_2 \circ f \circ \varphi_1^{-1}(w) = w^m S(w)$, with $S(0) \neq 0$ and S analytic. We can use a complex logarithm to write $S(w) = (h(w))^m$, hence

$$\varphi_2 \circ f \circ \varphi_1^{-1}(w) = \left(\underbrace{wh(w)}_{=: \psi(w)} \right)^m$$

but $\varphi'(0) = h(0) \neq 0$ thus by local inverse theorem φ has a holomorphic inverse near 0, hence

$$\varphi_2 \circ f \circ (\psi \circ \varphi_1)^{-1}(z) = z^m$$

and the chart is actually $\psi \circ \varphi_1$ on a subset of U_1 □

Definition 25

$m_P := \text{mult}_P(f)$ the **multiplicity**. If $m > 1$, then P is called a **ramification point**.

Proposition 26. *Ramification points corresponds to the set of vanishing points of $(\varphi_2 \circ f \circ \varphi_1^{-1})'$ at 0, hence it is in particular discrete.*

Theorem 27: Definition of degree

Let $f : X \rightarrow Y$, X, Y connected and f a proper nonconstant holomorphic map. Then, for all $y \in Y$,

$$d_y(f) = \sum_{f(x)=y} \text{mult}_x(f) \text{ is constant and finite}$$

It is denoted $\deg(f)$ and called the **degree** of f .

Proof. f being proper implies that $f^{-1}(\{y\})$ is compact. If it had infinitely many elements then I can construct a converging sequence. The local behavior on the accumulation point instead says that there should be only finitely many points in a neighborhood. Hence $f^{-1}(\{y\})$ is finite and thus also $d_y(f)$.

We want to show that $y \mapsto d_y$ is locally constant. Because X is connected, this would imply $y \mapsto d_y$ is constant. With this in mind, fix $y \in Y$ and pick a chart (V, ψ) at y . Let $f^{-1}(\{y\}) = \{x_1, \dots, x_n\}$. For all $i = 1 \dots n$ there exists $(U_i, \varphi_i) \ni x_i$ such that $\psi \circ f \circ \varphi_i^{-1}(z) = z^{m_i}$ and $m_i = \text{mult}_{x_i}(f)$. Now, for all $y' \in V$,

$$\sum_{x \in \bigcup_i^n U_i : f(x) = y'} \text{mult}_x(f) = d_{y'}(f) = \sum_{i=1}^n m_i$$

To show that $d_{y'}(f) = d_y(f)$, we need to check that

$$\{x \in X : f(x) = y'\} = \{x \in \bigcup_i^n U_i : f(x) = y'\}$$

. We show this by contradiction. Suppose indeed there exists some $(x'_k)_{k \in \mathbb{N}} \in f^{-1}(\{y'_k\}) \setminus \bigcup_i^n U_i$ for all $y'_k \rightarrow y$ as $k \rightarrow \infty$. By properness one can extract x'_k convergent.

□

Exercise 28 (T2 E1)

Soit $n \geq 1$. Montrer que la courbe projective $\mathcal{C} \subseteq \mathbb{P}^2(\mathbb{C})$ de Fermat d'équation homogène $x^n + y^n + z^n = 0$ est non singulière.

Since the equation is symmetric in the coordinates it suffices to check one affine chart, say $z \neq 0$. Hence $g(x, y) = x^n + y^n + 1 = 0$ in $\mathbb{C}^2$.

$$\frac{\partial g}{\partial x} = nx^{n-1} = 0 \implies x = 0 \quad ; \quad \frac{\partial g}{\partial y} = ny^{n-1} = 0 \implies y = 0$$

but $(0, 0) \notin \mathcal{C}$

Another possibility is using Euler's formula, and is left as an exercise.

Exercise 29 (T2 E2)

Consider $\mathcal{C} = \{y^2 = f(x)\} \subseteq \mathbb{C}^2$ where $f \in \mathbb{C}[x]$ of degree at least 3. Montrer les suivants:

- (1) affine curve $\mathcal{C}$ is nonsingular $\iff f$ doesn't have multiple roots
- (2) $\mathcal{C} \subseteq \mathbb{P}^2(\mathbb{C})$ is a projective curve with affine equation $y^2 = f(x)$
- (3) The point ∞ is nonsingular $\iff d = 3$

(1)

$$\begin{cases} \frac{\partial}{\partial x} : f'(x) = 0 \\ \frac{\partial}{\partial y} : 2y = 0 \implies y = 0 \end{cases}$$

if $(x, y) \in \mathcal{C}$ then $f(x) = 0$. Hence we get that $f(x) = f'(x) = 0$ but then this means that (x, y) is a singular point $\iff y = 0$ and x is a multiple root of f .

- (2) Homogenizing we consider $x = \frac{x_1}{x_0}$ and $y = \frac{x_2}{x_0}$ and we get that $\left(\frac{x_2}{x_0}\right)^2 = f\left(\frac{x_1}{x_0}\right) = \frac{1}{x_0^d} F(x_1, x_0)$ where F is the homogenization of f .

Now the equation becomes $x_0^{d-2} x_2 = F(x_1, x_0)$.

To now find the points at infinity consider $x_0 = 0$ thus $F(x_1, 0) = 0$ but $F(x_1, x_0) = a_d x_1^d + a_{d-1} x_1^{d-1} x_0 + \cdots + a_0 x_0^d$ hence $x_1 = 0$. We get that the only point at infinity is $[0 : 0 : 1] \in \mathcal{C}$

- (3) Let's look at the affine chart $\{x_2 \neq 0\}$. Let's call $u := \frac{x_1}{x_2}$ and $v := \frac{x_0}{x_2}$.

The equation in the coordinates (u, v) is $v^{d-2} = F(u, v)$. Let's check singularity:

$$\begin{cases} \frac{\partial}{\partial u} : \frac{\partial F}{\partial u} = 0 \\ \frac{\partial}{\partial v} : (d-2)v^{d-3} - \frac{\partial F}{\partial v} = 0 \end{cases}$$

Since F is homogenous, then its partial derivatives are also, hence $\frac{\partial F}{\partial u}(0, 0) = \frac{\partial F}{\partial v}(0, 0) = 0$. Then, if $d > 3$ $(0, 0)$ is a singular point and it is not if $d = 3$. Note that in these coordinates $(0, 0)$ is ∞ .

Exercise 30 (T2 E3)

Consider $a_0, \dots, a_n \geq 1$ integers. Consider the *projective space à la pold* the quotient $X :=$

$$\mathbb{P}(a_0, \dots, a_n) = (\mathbb{C}^{n+1} \setminus \{0\}) / \mathbb{C}^\times \text{ via the action } t(x_0, \dots, x_n) = (t^{a_0} x_0, \dots, t^{a_n} x_n)$$

In the following consider $a_0 = a_1 = \dots = a_n$

- (1) $U_0 = \{x_0 \neq 0\} \subseteq X$ is an affine chart and $U_0 \cong \mathbb{C}^n$ if $a_i = 1$ for all $i \geq 1$.

Exercise 31 (T2 E5)

Let w_1, w_2 a base of $\mathbb{C}$ as a real vector space. Let $f : \mathbb{C} \rightarrow \mathbb{C}$ be holomorphic such that $f(z + w_i) = \alpha_i f(z)$ for all $z \in \mathbb{C}$ and $i = 1, 2$.

Prove that $f(z) = \lambda e^{\mu z}$ for some $\lambda, \mu \in \mathbb{C}$

First note that

$$f(z + a_1 \omega_1 + a_2 \omega_2) = \alpha_1^{a_1} \alpha_2^{a_2} f(z) \quad a_1, a_2 \in \mathbb{Z}$$

we may suppose that $\alpha_1, \alpha_2 \neq 0$ since otherwise $f = 0$. Note then that

$$f'(z + \omega_i) = \alpha_i f'(z)$$

Let's then look at $F = f'/f$ (knowing that the answer we are trying to find is the exponential) which is invariant under Λ and meromorphic. If f is invertible, the F is an holomorphic elliptic function, hence is constant. But then clearly f is of the form $\lambda e^{\mu z}$ for some λ and μ .

We still need to show that $f \neq 0$. If $f(z_0) = 0$.

Corollary 32. Assume $f : X \rightarrow Y$ holomorphic between X, Y compact connected Riemann Surfaces. Suppose f is non constant. Then

$$\deg(f) = 1 \iff f \text{ is biholomorphic}$$

Proof. If the $\deg(f) = 1$ then it must be injective by definition of degree. $\square$

Example 33. If X is compact connected and $f : X \rightarrow \mathbb{C}$ is meromorphic with a unique simple pole. Then $X \cong \widehat{\mathbb{C}}$.

Indeed $f^{-1}(\{\infty\}) = \{P\}$ hence $\deg(f) = 1$ thus f is actually biholomorphic.

5.1 Δ -Complexes

It doesn't work that well, that is why generally singular homology is used.

Definition 34

A **simplex** is $\Delta^n = \{(x_0, \dots, x_n) \in \mathbb{R}^{n+1} : \sum_{i=0}^n x_i = 1, x_i \geq 0\}$. **faces** are $F_i = \{(\mathbf{x} \in \Delta^n : x_i = 0)\}$. There is a **face map** $d_i : \Delta^{n-1} \rightarrow \Delta^n$ mapping $(x_0, \dots, x_{n-1})$ to $(x_0, \dots, x_{i-1}, 0, x_i, \dots, x_{n-1})$. The **boundary map** is $\partial \Delta^n = \bigcup_{i=0}^n F_i$

Definition 35: Δ -complex

Let $(\Delta, X) = \{\sigma_\alpha : \Delta^{n_\alpha} \rightarrow X\}_{\alpha \in I}$ with I finite. These properties must be valid:

1. each map $\sigma_\alpha|_{\dot{\Delta}^{n_\alpha}}$ is injective and $X = \bigcup_{\alpha \in I} \sigma_\alpha(\dot{\Delta}^{n_\alpha})$
2. $\forall \alpha \in I$, and $\forall i$, $\sigma_\alpha \circ d_i : \Delta^{n_\alpha-1} \rightarrow X$ belong to (Δ, X)
3. The topology on X is such that $A \subseteq X$ is open if and only if $\forall \alpha \in I$, $\sigma_\alpha^{-1}(A) \subseteq \Delta^{n_\alpha}$ is open

Hence each $\sigma_\alpha : \Delta^{n_\alpha} \rightarrow \sigma_\alpha(\dot{\Delta}^{n_\alpha})$ is a homeo, and $X \cong \coprod_{\alpha \in I} \Delta^{n_\alpha} / \sim$ identification of faces.

Remark. the simplexes in the complex are not all the same.

Example 36. Let $\tilde{X} = \Delta^2 \times \{1, 2\}$ and $\pi_1 : \tilde{X} \rightarrow \Delta^2$ given by the projection $(x, i) \mapsto x$. Define next the relation

$$x \sim y \iff \exists (i, t) \in \{0, 1, 2\} \times \Delta^1 \text{ s.t. } \pi_1(x) = \pi_1(y) = d_i(t)$$

Finally $X = \tilde{X} / \sim$ and let's note $[x, i] \in X$ the equivalence class.

At this point there are the following

0-Simplexes: $\sigma_i^0 : e_i \mapsto [e_i, 1]$ for $i = 0, 1, 2$

1-Simplexes: $\sigma_i^1 : \Delta^1 \ni t \mapsto [d_i(t), 1]$ for $i = 0, 1, 2$

2-Simplexes: $\sigma_i^2 : \Delta^2 \ni t \mapsto [t, i]$ for $i = 1, 2$

5.1.1 Homology of Δ -simplexes

Consider the complex $(\Delta, X) = \{\sigma_\alpha\}_{\alpha \in I}$ and $n := \max_{\alpha \in I} n_\alpha < \infty$. For all $k = 0, \dots, n$ let C_k be the free abelian group generated by $\{\sigma_\alpha(\Delta^{n_\alpha}) : n_\alpha = k\}$.

Then the boundary map $\partial_k : C_k \rightarrow C_{k-1}$ is **defined** by

$$\partial_k \{\sigma_\alpha(\Delta^{n_\alpha})\} = \sum_{i=0}^k (-1)^i \sigma_\alpha \circ d_i(\Delta^{k-1})$$

Example 37. Let $\partial^0 = \{1\}$, $\partial^1 = \text{conv}(e_0, e_1) = [e_0, e_1]$ and $\Delta^2 = \text{conv}\{e_0, e_1, e_2\} = [e_0, e_1, e_2]$.

Hence, the boundary maps give $\partial_1 \Delta^1 = \{e_1\} - \{e_0\}$ and $\partial_2 \Delta^2 = [e_1, e_2] - [e_0, e_2] + [e_0, e_1]$

Proposition 38. *If we look at the sequence*

$$\{0\} \rightarrow C_n \xrightarrow{\partial_n} C_{n-1} \xrightarrow{\partial_{n-1}} \dots \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

is a complex of abelian groups, i.e. $\partial_{k-1} \circ \partial_k = 0$ for all k

Definition 39

We define the **homology groups**

$$H_k^\Delta = \frac{\text{Ker} \partial_k}{\text{Im} \partial_{k+1}}$$

Theorem 40

Δ -complex homology is isomorphic to *singular* homology.

Definition 41

The Euler characteristic

$$\chi_{(\Delta, X)} = \sum_{i=0}^n (-1)^i \text{rk}_{\mathbb{Z}}(H_i^\Delta) = \sum_{i=0}^n (-1)^i \text{rk}_{\mathbb{Z}}(C_i)$$

is a topological invariant

Theorem 42

- 1) All connected compact Riemann Surfaces are Δ -complexes with $n = 2$
- 2) If $\mathcal{F} \subseteq X$ is finite, one can choose (Δ, X) such that $\mathcal{F} \subseteq \bigcup_{n_\alpha=0} \sigma_\alpha(\Delta^0)$
- 3) $\chi_X = \sum_{i=0}^2 (-1)^i \text{rk}(C_i) = 2 - 2g$ is a topological invariant, where g is called *genus*.

Theorem 43: Riemann-Hurwitz

Let X, Y be compact connected Riemann surfaces and $f : X \rightarrow Y$ holomorphic non-constant. Then

$$2g(X) - 2 = \deg(f) (2g(Y) - 2) + \sum_{P \in X} (\text{mult}_P(f) - 1)$$

Proof. We want to avoid somehow the ramification points, since without them it would be a topological covering and much easier.

Let's call $R(f) := \{x \in X : \text{mult}_x(f) > 1\}$ and $B(f) := f(R(f))$. We know that $f : X \setminus f^{-1}\{B(f)\} \rightarrow Y \setminus B(f)$ is a topological covering of degree $\deg f$. Take

a Δ -complex on Y such that $B(f) \subseteq (\Delta, Y)_0$. Remark that $\overbrace{\sigma_\alpha(\Delta^2)}^o \subseteq (\Delta, Y)_2$ are simply connected.

Above each $\overbrace{\sigma_\alpha(\Delta^2)}^o$, f is a trivial covering. Denoting by $f_{j,\alpha}^{-1} : \overbrace{\sigma_\alpha(\Delta^2)}^o \rightarrow X$ all $d = \deg(f)$ local inverses of f .

The collection $\{f_{j,\alpha}^{-1} \circ \sigma_\alpha(\Delta^2)\}_{j=1..d}$ can be completed as a Δ -complex on X . In particular, one need to add 0-simplexes and 1-simplexes. We note that $f^{-1}\{B(f)\} \subseteq (\Delta, X)_0$.

By the covering property, $\#(D, X)_2 = d\#(\Delta, Y)_2$ and similarly $\#(\Delta, X)_1 = d\#(\Delta, Y)_1$.

The part to count are $\#(\Delta, X)_0$. Pick $q \in (\Delta, Y)_0$ and

$$\#f^{-1}(\{q\}) = \sum_{f(p)=q} 1 = \sum_{f(p)=q} (1 - \text{mult}_p(f) + \text{mult}_p(f)) = d + \sum_{f(p)=q} (1 - \text{mult}_p(f))$$

Hence

$$\#(\Delta, X)_0 = \sum_{q \in (\Delta, Y)_0} \#f^{-1}(\{q\}) = d\#(\Delta, Y)_0 + \sum_{q \in (\Delta, Y)_0} \sum_{f(p)=q} (1 - \text{mult}_p(f))$$

but the last double sum is just $\sum_{p \in X} (1 - \text{mult}_p(f))$. Finally,

$$\chi(X) = \#(\Delta, X)_2 - \#(\Delta, X)_1 + \#(\Delta, X)_0 = d\chi(Y) + \sum_{P \in X} (1 - \text{mult}_p(f))$$

□

5.1.2 Applications of R.H.

Theorem 44: Placker's formula

let $\mathcal{C}$ be a (smooth) projective curve of degree d . then

$$g(\mathcal{C}) = \frac{(d-1)(d-2)}{2}$$

proof sketch. Consider $\mathcal{C} = \{(x, y, z) \in \mathbb{P}^2\mathbb{C} : F(x, y, z) = 0\}$ and the projection $\pi : \mathcal{C} \rightarrow \mathbb{P}^1\mathbb{C} \cong \widehat{\mathbb{C}}$ given by $\pi([x : y : z]) = [x : z]$. Then π is a holomorphic map of degree d .

Ramification points are the points $\{[x : y : z] \in \mathcal{C} : \frac{\partial F}{\partial y} = 0\}$. Then from R.H. (theorem 5.1.1) we get that

$$2g(\mathcal{C}) - 2 = d(-2) + \sum_{x \in \mathcal{C}} (\text{mult}_x(\pi) - 1) \quad (5.1.1)$$

Then, by Bézout (a geometric algebra theorem),

$$\sum_{x \in \mathcal{C}} (\text{mult}_x(\pi) - 1) = \deg(F) \cdot \deg\left(\frac{\partial F}{\partial y}\right) = d(d-1) \quad (5.1.2)$$

By equations (5.1.1) and (5.1.2) we get that

$$g(\mathcal{C}) = \frac{1}{2}(2 - 2d + d(d-1)) = \frac{(d-1)(d-2)}{2}$$

□

Chapter 6

Differential forms and Stokes

Notations Some common notations in this chapter are, if $U \subseteq \mathbb{C}$ is an open set, $f \in C^1(U)$, then

$$\frac{\partial f}{\partial z} := \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right) \quad ; \quad \frac{\partial f}{\partial \bar{z}} := \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right) \quad ; \quad df := \frac{\partial f}{\partial z} dz + \frac{\partial f}{\partial \bar{z}} d\bar{z}$$

If $\varphi : V \rightarrow U$ is holomorphic, then

$$\frac{\partial f \circ \varphi}{\partial z} = \varphi'(z) \left(\frac{\partial f}{\partial z} \circ \varphi \right) \quad ; \quad \frac{\partial f \circ \varphi}{\partial \bar{z}} = \overline{\varphi'(z)} \left(\frac{\partial f}{\partial \bar{z}} \circ \varphi \right)$$

Definition 45

A 1-form on U is $\omega = p(z) dz + q(z) d\bar{z}$ and

$$\varphi^* \omega \stackrel{\text{def}}{=} p(\varphi(z)) \varphi'(z) dz + q(\varphi(z)) \overline{\varphi'(z)} d\bar{z}$$

Remark.

$$d(f \circ \varphi) = \varphi^*(df)$$

Definition 46

Let $(U_\alpha, \varphi_\alpha)$ be an atlas on X . Then

- A 1-form ω is a collection ω_α of 1-form on $\varphi_\alpha(U_\alpha) \subseteq \mathbb{C}$
- $\omega_{\alpha_2} = (\varphi_{\alpha_1} \circ \varphi_{\alpha_2}^{-1})^* \omega_{\alpha_1}$ whenever $U_{\alpha_1} \cap U_{\alpha_2} \neq \emptyset$

Such that ω is well defined on X .

Definition 47: 2-forms

If $\eta = f(z) dz \wedge d\bar{z}$ (on $U \subseteq \mathbb{C}$) and $\varphi : V \rightarrow U$ is *holomorphic*, then

$$\varphi^* \eta = f \circ \varphi |\varphi'(z)|^2 dz \wedge d\bar{z}$$

Remark. Let $\omega = p dz + q d\bar{z}$ be a 1-form. Then

$$d(\varphi^* \omega) = \left(\frac{\partial \varphi}{\partial z} \circ q - \frac{\partial p}{\partial \bar{z}} \circ \varphi \right) |\varphi'|^2 dz \wedge d\bar{z} \implies d(\varphi^* \omega) = \varphi^*(d\omega)$$

Definition 48: 2-forms

Let $(U_\alpha, \varphi_\alpha)$ be an atlas on X . Then

- A 2-form η is a collection η_α of 2-forms on $\varphi_\alpha(U_\alpha) \subseteq \mathbb{C}$
- $\eta_{\alpha_2} = (\varphi_{\alpha_1} \circ \varphi_{\alpha_2}^{-1})^* \eta_{\alpha_1}$ whenever $U_{\alpha_1} \cap U_{\alpha_2} \neq \emptyset$

Such that η is well defined on X .

We will denote by

- $\Omega^0(X)$ the set of C^∞ functions on X
- $\Omega^1(X)$ the set of C^∞ 1-forms on X
- $\Omega^2(X)$ the set of C^∞ 2-forms on X

The operator d is well defined

$$\Omega^0(X) \xrightarrow{d} \Omega^1(X) \xrightarrow{d} \Omega^2(X)$$

and moreover $d \circ d = 0$

Formulas: if I take $f \in \Omega^0(X)$, then I can set locally $\partial f = \frac{\partial f}{\partial z} dz$ and $\bar{\partial} f = \frac{\partial f}{\partial \bar{z}} d\bar{z}$. Similarly, if $\omega = p dz + q d\bar{z}$ is a 1-form, then I can set locally

$$\partial \omega = \frac{\partial q}{\partial z} dz \wedge d\bar{z} \quad ; \quad \bar{\partial} \omega = -\frac{\partial p}{\partial \bar{z}} dz \wedge d\bar{z}$$

So we have $d = \partial + \bar{\partial}$ and $d^2 = 0 = \bar{\partial}^2$.

For 1-forms, we have a decomposition. Let us denote by $\Omega^{(1,0)}(X)$ the set of 1-forms such that $\omega = p(z)dz$ locally and similarly by $\Omega^{(0,1)}(X)$ the set of 1-forms such that locally $\omega = q(z)d\bar{z}$. If p is holomorphic (meromorphic), then ω is called holomorphic (meromorphic).

The following diagram holds:

$$\begin{array}{ccccc} & & \Omega^0(X) & & \\ & \swarrow \partial & & \searrow \bar{\partial} & \\ \Omega^{(1,0)}(X) & & & & \Omega^{(0,1)}(X) \\ & \searrow \bar{\partial} & & \swarrow \partial & \\ \partial \downarrow & & \Omega^2(X) & & \downarrow \bar{\partial} \\ \{0\} & & & & \{0\} \end{array}$$

6.1 Integration and differential forms

Let $\omega \in \Omega^1(X)$ and $C : [0, 1] \rightarrow X$ be smooth. We want to get to a sensible definition of $\int_C \omega$.

Pick a partition $0 = t_0 < t_1 < \dots < t_n = 1$ of $[0, 1]$ such that $\forall k = 0, \dots, n-1$ we have $C([t_k, t_{k+1}]) \subseteq (U_k, \varphi_k)$ a chart. At this point we can define

$$\int_C \omega = \sum_{k=0}^{n-1} \underbrace{\int_{\varphi_k(C[t_k, t_{k+1}])}}_{\text{curve in } \mathbb{C}} \overbrace{(\varphi_k^{-1})^* \omega}^{\text{1-form on } \varphi_k(U_k) \subseteq \mathbb{C}} \quad (6.1.1)$$

which in particular *does not* depend on *charts* and *subdivision*.

In **local coordinates**, we have

$$\int_{\gamma} (p dx + \varphi dy) \stackrel{\text{def}}{=} \int_0^1 ((p \circ \gamma(t))\gamma_1'(t) + (q \circ \gamma(t))\gamma_2'(t)) dt$$

where $\gamma = (\gamma_1, \gamma_2) : [0, 1] \rightarrow \mathbb{R}^2$.

For 2-forms it's a bit harder.

Let $K \subseteq X$ be *compact*, $\eta \in \Omega^2(X)$ with $(U_i, \varphi_i)_{i \in I}$ a locally finite atlas on X .

Let $(p_i)_{i \in I}$ be a *partition of unity* such that

1. $\forall i, p_i \in C^\infty(X)$, $\text{supp}(p_i) \subseteq U_i$ and $p_i \geq 0$
2. $\forall x \in X, \sum_{i \in I} p_i(x) = 1$

Then we can define

$$\int_K \eta = \sum_{i \in I} \int_{\varphi_i(K \cap U_i)} (p_i \circ \varphi_i^{-1})(\varphi_i^{-1})^* \eta$$

which does not depend on $(p_i)_{i \in I}$ or $(U_i, \varphi_i)_{i \in I}$.

Let $K \subseteq X$ compact such that $\mathring{K} \neq \emptyset$ and ∂K is a finite disjoint union of simple loops, i.e. ∂K is a 1-dimensional submanifold of X . We will call all of this being a *nice* compact set.

Proposition 49. *Riemann Surfaces are orientable*

Hence ∂K is naturally oriented.

Theorem 50: Stokes

Let $K \subseteq X$ *nice* as in the previous paragraph and $\omega \in \Omega^1(X)$, then

$$\int_K d\omega = \int_{\partial K} \omega$$

Corollary 51 (Consequences of Stokes). *Let X compact, $f \in \Omega^0(X)$ and $\omega \in \Omega^1(X)$. Then*

- if $\omega \in \Omega^{(0,1)}(X)$, then $\int_X f \partial\omega = - \int_X \partial f \wedge \omega$
- if $\omega \in \Omega^{(1,0)}(X)$, then $\int_X f \bar{\partial}\omega = - \int_X \bar{\partial} f \wedge \omega$

Proof. In general, $d(f\omega) = (df) \wedge \omega + f d\omega$. Integrating, we get

$$\begin{aligned} \int_X f d\omega &= \underbrace{\int_X d(f\omega)}_{= 0 \text{ by Stokes } (\partial X = \emptyset)} - \int_X (df) \wedge \omega \end{aligned}$$

and if $\omega \in \Omega^{(0,1)}(X)$, then $d\omega = \partial\omega$ and (not obvious) $(df) \wedge \omega = \partial f \wedge \omega$. Analogously with $\omega \in \Omega^{(1,0)}(X)$. □

Definition 52: Laplacian

Let $f \in \Omega^0(X)$. Then the **laplacian**

$$\Delta f := 2i\bar{\partial}\partial f = -2i\partial\bar{\partial}f \in \Omega^2(X)$$

which in local charts is

$$\Delta f = -\left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}\right) dx \wedge dy$$

Note. Δf is a 2-form, not a function

Which leads to this rephrasing of Stokes:

Theorem 53: Green's formula

For all $f, g \in \Omega^0(X)$ (X compact), we have

$$\int_X (\Delta f)g = \int_X f \Delta g = -\frac{1}{2i} \int_X \bar{\partial} f \wedge \partial g$$

Proof. Integration by parts. $\square$

Corollary 54. Assume X compact connected. Let $f \in \Omega^0(X)$ be such that $\Delta f = 0$. Then f is constant.

Proof. Integrating $(\Delta f)\bar{f}$ we get

$$0 = \int_X (\Delta f)\bar{f} = -\frac{1}{2i} \int_X \bar{\partial} f \wedge \partial \bar{f} = \frac{1}{2i} \int_X \partial f \wedge \bar{\partial} \bar{f}$$

and in local coordinates, $\bar{\partial} f \wedge \partial \bar{f} = \left| \frac{\partial f}{\partial \bar{z}} \right|^2 d\bar{z}$. Since $\int_X (\Delta f)\bar{f}$ is a sum of positive terms, we get that locally $\frac{\partial f}{\partial \bar{z}} = 0$. Hence, f is holomorphic, thus constant. $\square$

6.1.1 Meromorphic 1-forms

Definition 55: Residue

Let ω be a meromorphic 1-form on X . Let p be a pole and (U, φ) be a chart such that p is the only pole in U . Then we set

$$\text{Res}_p(\omega) := \frac{1}{2\pi i} \int_{\gamma_p} (\varphi^{-1})^* \omega$$

where γ_p is a simple loop (with positive orientation) around $\varphi(p)$ in $\varphi(U) \subseteq \mathbb{C}$.

Proposition 56 (Bonne definition). *Considering the previous definition,*

1. *It does not depend on (U, φ) and γ_p .*
2. *If in (U, φ) , $(\varphi^{-1})^* \omega = \left(\frac{a_0}{z} + \dots + \frac{a_k}{z^k} + H(z) \right) dz$ where H is holomorphic, then $\text{Res}_p(\omega) = a_0$*

Theorem 57: Residue

Let ω be a meromorphic 1-form on X compact. Then

$$\sum_{p \text{ pole}} \text{Res}_p(\omega) = 0$$

Proof. Let $p_1, \dots, p_n$ be all of the (distinct) poles of ω . pick charts $(U_k, \varphi_k) \ni p_k$ such that $\overline{U_i} \cap \overline{U_j} = \emptyset$ for all $i \neq j$. Assume that ∂U_k is a simple loop around p_k . Then

$$X \setminus \bigcup_{i=1}^n U_i =: K \quad \text{is nice, hence by Stokes} \quad \int_K d\omega = \int_{\partial K} \omega$$

But $d\omega = \bar{\partial}\omega = 0$ since ω is **holomorphic** on K . It follows that

$$0 = \int_{\partial K} \omega = - \sum_{i=1}^n \int_{\partial U_i} \omega = - \sum_{i=1}^n \text{Res}_{p_i}(\omega)$$

□

Corollary 58. *Let f be a non-constant meromorphic function on X compact. Then $\#Z(f) - \#P(f) = 0$ where $Z(f)$ is the set of zeros of f and $P(f)$ is the set of poles of f*

Proof. Consider $\frac{\partial f}{f}$ which is a meromorphic 1-form. By the residue theorem, $\sum_p \text{Res}_p(\omega) = 0$. Then poles of ω come from poles and zeros of f . Indeed if x is a zero, then $\text{Res}_x \omega = \text{mult}_x(f)$ and if p is a pole, then $\text{Res}_p \omega = -\text{mult}_p(f)$ □

Theorem 59: Degree of the canonical divisor

Let $\omega \neq 0$ be a 1-form, then

$$\#Z(\omega) - \#P(\omega) = 2g(X) - 2$$

where $Z(\omega)$ is set of zeros, $P(\omega)$ is the set of poles and $g(X)$ is the genus of X .

Proof sketch. This is an application of Theorem 5.1.1.

1. If ω, ω' are two meromorphic 1-forms, say $\omega = P(z)dz$ and $\omega' = Q(z)dz$ locally. Set $f = P(z)/Q(z)$, then when changing coordinates

$$f(\varphi(z)) = \frac{P(\varphi(z))}{Q(\varphi(z))} = \frac{P(z)\varphi'(z)}{Q(z)\varphi'(z)} = \frac{P(z)}{Q(z)}$$

making f well-defined. Hence, $\omega = f\omega'$ where f is a well defined meromorphic function on X . Hence

$$\#Z(\omega) - \#P(\omega) = \underbrace{\#Z(f) - \#P(f)}_{=0 \text{ by Corollary 58}} + \#Z(\omega') - \#P(\omega')$$

2. We use the fact that on any X there exists a non trivial meromorphic function $h : X \rightarrow \widehat{\mathbb{C}}$ (holomorphic). On $\widehat{\mathbb{C}}$, dz has a double pole at ∞ and is nonvanishing.
3. Consider $\omega = h^*(dz)$. Note that the zeros of ω are exactly the ramification points of h , and have order $\text{mult}_x(h) - 1$. As for the poles, $P(\omega) = h^{-1}(\{\infty\})$ with multiplicity $2\text{mult}_x(h)$, whenever $h(x) = \infty$. At this point, we can apply the Riemann Hurwitz formula (theorem 5.1.1) obtaining

$$\begin{aligned} \#Z(\omega) - \#P(\omega) &= \sum_{p \in X} (\text{mult}_p(h) - 1) - \overbrace{2}^{=\chi(\widehat{\mathbb{C}})} \underbrace{\deg(h)}_{\sum_{x \in h^{-1}(\infty)} \text{mult}_x(h)} = \\ &\stackrel{R.H.}{=} -\chi(X) = 2g(X) - 2 \end{aligned}$$

□

Chapter 7

Cohomology and Serre duality

Before approaching this chapter, it's useful to remind all of the notations that will be used, introduced in Chapter 6.

- $\Omega^0(X)$ the set of C^∞ functions on X
- $\Omega^1(X)$ the set of C^∞ 1-forms on X
- $\Omega^2(X)$ the set of C^∞ 2-forms on X
- $\Omega^{(1,0)} = \{\omega \in \Omega^1(X) : \omega = P(z) dz\}$
- $\Omega^{(0,1)} = \{\omega \in \Omega^1(X) : \omega = Q(z) d\bar{z}\}$

Finally, the following diagram holds:

$$\begin{array}{ccccc}
 & & \Omega^0(X) & & \\
 & \swarrow \partial & & \searrow \bar{\partial} & \\
 \Omega^{(1,0)}(X) & & & & \Omega^{(0,1)}(X) \\
 \downarrow \partial & \searrow \bar{\partial} & & \swarrow \partial & \downarrow \bar{\partial} \\
 \{0\} & & \Omega^2(X) & & \{0\}
 \end{array}$$

7.1 De Rham Cohomology

Let X be a connected compact Riemann Surface. Then The following sequence is a complex

$$\{0\} \hookrightarrow \Omega^0(X) \xrightarrow{d_1} \Omega^1(X) \xrightarrow{d_2} \Omega^2(X) \rightarrow \{0\} \quad (\text{as } \mathbb{C}\text{-vector spaces})$$

And it leads to the following cohomology spaces

- $H_{DR}^0 = \{f \in \Omega^0(X) : df = 0\} \cong \mathbb{C}$
- $H_{DR}^1 = \{\omega \in \Omega^1(X) : d\omega = 0\} / \{df : f \in \Omega^0(X)\}$
- $H_{DR}^2 = \{\eta \in \Omega^2(X)\} / \{d\omega : \omega \in \Omega^1(X)\}$

Theorem 60: De Rham for Surfaces

It holds that

$$H_{DR}^0 \cong H_{DR}^2 \cong \mathbb{C} \quad \text{and} \quad \mathbb{H}_{DR}^1 \cong \mathbb{C}^{2g}$$

where g is the genus of X

7.2 Dolbeault Cohomology

The cohomology spaces are

- $H^{(1,0)}(X) = \{\omega \in \Omega^{(1,0)}(X) : \bar{\partial}\omega = 0\}$ which is exactly the set of holomorphic 1-forms
- $H^{(0,1)}(X) = \{\omega \in \Omega^{(0,1)}(X)\} / \{\partial f : f \in \Omega^0(X)\}$

And there are some observations to make.

1. $H^{(1,0)}(X) \hookrightarrow H_{DR}^1(X)$. Indeed, if ω is a holomorphic 1-form and if $\omega = df$, then in local coordinates

$$\omega = P(z) dz = \frac{\partial f}{\partial z} dz + \underbrace{\frac{\partial f}{\partial \bar{z}}}_{=0} d\bar{z} \implies f \text{ is holomorphic} \implies f \text{ is constant}$$

hence $\omega = df = 0$

2. Similarly, setting $\overline{H^{(1,0)}}(X) = \{\omega \in \Omega^{(0,1)}(X) : \partial\omega = 0\}$ the set of anti-holomorphic 1-forms, then we also have $\overline{H^{(1,0)}}(X) \hookrightarrow H_{DR}^1(X)$ which is shown using the same logic.

Indeed, there is a stronger relationship

Theorem 61

The following equivalences hold:

$$H_{DR}^1(X) = H^{(1,0)}(X) \oplus \overline{H^{(1,0)}}(X) \quad (7.2.1)$$

$$\dim H^{(1,0)}(X) = \dim \overline{H^{(1,0)}}(X) = g \quad (7.2.2)$$

Lemma 62. Assume $\eta \in \Omega^2(X)$ such that $\int_X \eta = 0$. Then there exists $f \in C^\infty(X)$ such that $\Delta f = \eta$

The proof is articulated in a *soft proof* with techniques used in Hilbert spaces (Lax-Milgram) and an argument about local ellipticity.

Another possibility is to use PDEs.

End of proof. Fix $\omega \in \Omega^1(X)$ such that $d\omega = 0$. We look for $\omega = df + h + g$, where h is holomorphic and g anti-holomorphic. Say $\omega = \omega^{(1,0)} + \omega^{(0,1)}$. Since $d\omega = 0$, then $\bar{\partial}\omega^{(1,0)} = -\partial\omega^{(0,1)}$. Let $f \in C^\infty(X)$ and write

$$\omega - df = \underbrace{(\omega^{(1,0)} - \partial f)}_{=:h} + \underbrace{(\omega^{(0,1)} - \bar{\partial} f)}_{=:g}$$

and we want to have h holomorphic and g anti-holomorphic. However, $\bar{\partial}h = 0 \iff \bar{\partial}\omega^{(1,0)} - \bar{\partial}\partial f = 0 \iff \partial\omega^{(0,1)} - \partial\bar{\partial}f = 0$, hence all is left is to solve

$$\bar{\partial}\omega^{(1,0)} = \bar{\partial}\partial f$$

But

$$\int_X \bar{\partial}\omega^{(1,0)} = \int_X d\omega^{(1,0)} = 0 \text{ by Stokes}$$

Hence $\exists f$ such that $\Delta f = \bar{\partial}\omega^{(1,0)}$ by solving Poisson's equation $\square$

Proof. The proof starts by constructing the isomorphism $\omega \mapsto \bar{\omega}$ which maps $\omega \in H^{(1,0)}$ with $\omega = P(z) dz$, to $\bar{\omega} = \bar{P}(z) d\bar{z} \in \overline{H^{(1,0)}}(X)$. It's easy to show that this is an anti-linear isomorphism, reaching the second equation as a consequence.

In conclusion, we have $\dim H^{(1,0)}(X) = g$ $\square$

Proposition 63. On $H^{(1,0)}(X)$,

$$\langle \omega_1, \omega_2 \rangle := -\frac{1}{2i} \int_X \omega_1 \wedge \bar{\omega}_2$$

is an inner product

Proof. To show that is *positive definite*, note that locally $-\frac{1}{2i}\omega \wedge \bar{\omega} = |P|^2 dx \wedge dy$, hence $\langle \omega, \omega \rangle = 0$ implies $\omega = 0$. Everything else is obvious. $\square$

7.2.1 Serre's Duality

Theorem 64

$$[\beta] = 0 \text{ in } H^{(0,1)}(X) \iff \forall \alpha \in H^{(1,0)}(X), \int_X \alpha \wedge \beta = 0$$

Corollary 65.

$$H^{(0,1)}(X) \cong \left(H^{(1,0)}(X)\right)^* \quad \text{and} \quad \dim H^{(0,1)}(X) = g$$

Proof of corollary. Note that

$$\begin{aligned} \zeta : H^{(0,1)} &\longrightarrow \left(H^{(1,0)}\right)^* \\ [\beta] &\longmapsto \zeta([\beta]) = \left(L_\beta : \alpha \mapsto \int_X \alpha \wedge \beta\right) \end{aligned}$$

By the theorem, ζ is injective. By Riesz on $(H^{(1,0)}, \langle \cdot, \cdot \rangle)$, for any $\varphi \in (H^{(1,0)}(X))^*$, there exist $\omega_0 \in H^{(1,0)}$ such that

$$\varphi(\omega) = \langle \omega, \omega_0 \rangle = -\frac{1}{2i} \int_X \omega \wedge \bar{\omega}_0 = \int_X \omega \wedge \left[-\frac{1}{2i} \bar{\omega}_0\right]$$

$\square$

Proof of Theorem 7.2.1. $\implies$ Obvious by Stokes

$\Leftarrow$ Let $[\beta] \in H^{(0,1)}(X)$ such that $\forall \alpha \in H^{(1,0)}(X), \int_X \alpha \wedge \beta = 0$. Note that $d\beta = \partial\beta$, hence $\int_X \partial\beta = 0$, which implies that $\exists f \in C^\infty(X)$ such that $\partial\bar{\partial}f = \partial\beta$, hence $\bar{\partial}(\bar{\partial}f - \beta) = 0$ and thus $\bar{\partial}(\bar{\beta} - \partial\bar{f}) = 0$. Note that the argument $\bar{\beta} - \partial\bar{f} \in H^{(1,0)}(X)$. We conclude

$$\|\bar{\beta} - \partial\bar{f}\|^2 = \int_X (\bar{\beta} - \partial\bar{f}) \wedge (\beta - \bar{\partial}f) = \int_X (\bar{\beta} - \partial\bar{f}) \wedge \beta = 0$$

hence $\bar{\beta} - \partial\bar{f} = 0$ which implies that $\bar{\partial}f = \beta$ $\square$

7.2.2 Application: Existence of meromorphic functions

Theorem 66

Let X be a compact connected Riemann Surface. Let $p \in X$. Then there exists f meromorphic function on X non-constant, such that f has a single pole at p with $\text{mult}_p(f) \leq g + 1$

Proof. Let $(U, \varphi) \ni p$. Let $q_0 \in C^\infty(X)$ such that $\text{Supp}(q_0) \subseteq U$ and $q_0 \equiv 1$ on $p \in \bar{V} \subseteq U$ and V open.

For all $\mathbf{a} = (a_1, \dots, a_{g+1}) \in \mathbb{C}^{g+1}$, set

$$g_{\mathbf{a}}(z) := q_0(z) \left(\sum_{k=1}^{g+1} \frac{a_k}{[\varphi(z)]^k} \right)$$

and extend it by 0 outside U .

Then $g_{\mathbf{a}}$ is meromorphic on V , but not on X . However,

$$\bar{\partial} g_{\mathbf{a}} = (\bar{\partial} q_0) \sum_{k=1}^{g+1} \frac{a_k}{[\varphi(z)]^k}$$

and this is all equal to 0 on V , since $g_{\mathbf{a}} \equiv 1$. This means that $\bar{\partial} g_{\mathbf{a}} \in \Omega^1(X)$.

The map

$$\begin{aligned} \mathbb{C}^{g+1} &\longrightarrow H^{(0,1)}(X) \\ \mathbf{a} &\longmapsto [\bar{\partial} g_{\mathbf{a}}] \end{aligned}$$

is a linear map, and $\dim H^{(0,1)}(X) = g < g + 1$, hence there exists $\mathbf{a} \in \mathbb{C}^{g+1} \setminus \{0\}$ such that $[\bar{\partial} g_{\mathbf{a}}] = 0$, hence there exists $f \in C^\infty(X)$ such that $\bar{\partial} g_{\mathbf{a}} = \bar{\partial} f$

Now observe that $\mathfrak{f} := g_{\mathbf{a}} - f$ is holomorphic on $X \setminus \{p\}$ ($\bar{\partial} \mathfrak{f} = 0$). Near p (on V),

this can be expressed as $\underbrace{\mathfrak{f}(z)}_{\bar{\partial}=0} = \underbrace{\sum_{k=1}^{g+1} \frac{a_k}{[\varphi(z)]^k}}_{\bar{\partial}=0} + \underbrace{f(z)}_{\text{holo near } p}$. Hence, $\mathfrak{f}$ is meromorphic. □

Corollary 67. *If X is a compact connected Riemann Surface and $g(X) = 0$, then $X \cong \hat{\mathbb{C}}$ as Riemann Surfaces.*

Proof. Since $g = 0$, there exists a meromorphic map with a single simple pole $f : X \rightarrow \hat{\mathbb{C}}$. Hence $\deg f = 1$, then f is an isomorphism. □

Exercise 68 Variant

Let X compact connected Riemann Surface with genus g . Show that given $p_1, \dots, p_{g+1}$ different points on X , then there exists a meromorphic function such that the set of poles is a subset of $\{p_1, \dots, p_{g+1}\}$ and all poles are simple.